Junyoung Park

Master Student

(+82) 10-3662-5451 | Seoul, Republic of Korea | pjy5451@gmail.com | linkedin.com/in/junyoungp
Chung-Ang University | 84 Heukseok-Ro, Dongjak-Gu, Seoul, Korea 06974

Research Interests

My research is at the intersection of various engineering disciplines, including structural dynamics, computer science, and electronics. I am focused on enhancing infrastructure safety and efficiency through a multidisciplinary approach. Specially, I employ IoT sensor, cloud computing, deep learning, and insights from structural dynamics to assess and continuously monitor the safety of bridges.

Mainly my research is dedicated towards achieving long-term bridge monitoring by developing high-resolution IoT sensors that measure acceleration and strain, called JANET. This sensor can perform continuous measurements for over 24 hours and are designed to either trigger data collection through excessive vibrations or record data at specific time intervals, ensuring long-term monitoring without the need for frequent battery replacements. Additionally, I invest significant effort in analyzing the collected data to derive various indicators such as natural frequencies, vibration usability, and displacement from data fusion, all of which contribute to assessing the safety of bridges. Further, my research extends to not only bridges but also the real-time monitoring of prefabricated structures throughout their lifecycle and monitoring of harbor infrastructures to ensure the highest standards safety and functionality.

Ultimately, my goal is to pave the way for smart infrastructure management, where data-driven insights and advanced technologies converge to enable the proactive an intelligent maintenance of critical structures. By combining various cutting-edge technologies, my research aims to make our infrastructure safer, reliable, and sustainable.

Education

Master / ***Chung-Ang University* – Seoul, Korea** Mar 2022 – Feb 2024

Master of Engineering (Major: Structural Engineering)
• Thesis: A Scalable Bridge Health Monitoring System using an IoT Sensor and Cloud Computing (In-progress)
• GPA 4.38/4.5

Bachelor / ***Chung-Ang University* – Seoul, Korea** Mar 2017 – Feb 2022

Double Degree in Bachelor of Engineering in Civil Engineering and Bachelor of Science in Mathematics
• Thesis: Development of Low-power IoT Sensor and a Cloud-based Data Fusion Displacement Estimation Method for Ambient Bridge Monitoring
• GPA 4.20/4.5

Honors & Awards

Gold Award / ***National Research Foundation of Korea*,** The 2nd X-Corps Plus Festival Nov 2022

CAU GRS Scholarship / ***Chung-Ang University*,** Received full funding during Graduate studies Mar 2022

Excellence Award / ***Korea Land and Housing Corporation*,** The 12th LH Land Development Technology Contest Nov 2021

Innovation Award / ***Ministry of Land, Infrastructure and Transport*,** Smart Construction Challenge 2021 Nov 2021

SeAH Haiam Scholarship / ***SeAH Haiam Scholarship Foundation*,** Granted 100% tuition fee waiver during 2 years of undergraduate study (3rd grade & 4th grade) Jun 2020

Minister's Award / ***Ministry of the Interior and Safety*,** Smart Earthquake Detection Competition Feb 2019

Grand Prize / ***Korea Institute for Advancement of Technology*,** 2018 Engineering Festival – AI Object Recognition Hackathon Nov 2018

Research Experience

Project-1 / IoT-based Smart Earthquake Detection 2019 - 2020

• Development of IoT sensor for seismic detection, interfacing an MCU with beacon and an accelerometer
• Proposed a 2-stage algorithm for event detection based on STA/LTA and frequency-based earthquake detection

Project-2 / Smart Ultra-High Performance Concrete with Self-Stress Sensing 2019 - 2020

• Designed and fabricated a smart concrete circuit for self-stress measurement
• Developed a beacon-based data collection pipeline with 1:N data collection capability

Project-3 / Smart Bridge using IoT Sensors and Cloud-based Analytics 2020 - Present

• Designed a circuit for a stand-alone, event-driven system for long-term vibration and timer-based monitoring
• Developed IoT sensor for simultaneous acceleration and multi-channel strain measurement, transmitting data to a cloud server via LTE
• Designed a bridge long-term data management framework based on MySQL and InfluxDB
• Designed a sensor-data analysis pipeline for bridge substructure evaluation

Project-4 / Prefab Structure Monitoring from Fabrication to Construction 2020 - Present

• Designed and developed an ultra-low Power, multi-channel strain visualization sensor using buried strain gauges for monitoring the entire lifecycle of prefabricated structures
• Optimized a QR code display algorithm for more efficient data storage on a E-paper
• Implemented a QR code data acquisition method through Kakaotalk along with analysis reporting using the ChatGPT API

Project-5 / Prefab Structure Monitoring during Transportation 2020 - Present

• Developed a high-resolution IoT sensors for simultaneous monitoring of acceleration and deformation of prefabricated structures during transportation
• Designing a gateway PCB for collecting 1:N Bluetooth communication measured data from IoT sensors and transmitting it to a server via LTE
• Integrated an automatic analysis system using the vibration, inclination, and deformation data generated during lifting, transportation, and loading processes

Project-6 / Smart Maintenance for Port Structure 2021 - Present

• Developed sensors for impact detection and long-term behavior monitoring of port structures based on ship berthing/unberthing
• Designed a data pipeline for real-time data collection using MQTT and data storage in InfluxDB

Skills

- **Programming Languages:** C, Python, MATLAB, node.js, swift
- **Development Tools:** MATLAB(Simulink), Altium Designer, Solidworks, PlatformIO, Xcode
- **Soft Skills:** Responsibility, Leadership, Critical Thinking
- **Languages:** English (Proficient), Korean (Native)

JOURNAL PUBLICATIONS

- Won, J., **Park, J.**, Park, J. W., & Kim, I. H. (2020). BLESeis: low-cost IOT sensor for smart earthquake detection and notification. *Sensors*, *20*(10), 2963.
- Khan, S., Won, J., Shin, J., **Park, J.**, Park, J. W., Kim, S. E., ... & Kim, D. J. (2021). SSVM: An ultra-low-power strain sensing and visualization module for long-term structural health monitoring. *Sensors*, *21*(6), 2211.
- Won, J., Park, J. W., **Park, J.**, Shin, J., & Park, M. (2021). Development of a reference-free indirect bridge displacement sensing system. *Sensors*, *21*(16), 5647.
- Le, H. V., Kim, T. U., Khan, S., **Park, J. Y.**, Park, J. W., Kim, S. E., ... & Kim, D. J. (2021). Development of low-cost Wireless sensing System for Smart ultra-high performance concrete. *Sensors*, *21*(19), 6386.
- Khayam, S. U., Won, J., Shin, J., **Park, J.**, & Park, J. W. (2023). Monitoring Precast Structures during Transportation Using a Portable Sensing System. *Automation in Construction*, *145*, 104639.
- Khayam, S. U., Ajmal, A., **Park, J.**, Kim, I. H., & Park, J. W. (2023). Tendon Stress Estimation from Strain Data of a Bridge Girder Using Machine Learning-Based Surrogate Model. *Sensors*, *23*(11), 5040.

CONFERENCE PROCEEDINGS

- Jongbin Won, **Junyoung Park**, Jong-Woong Park, "BLESeis: Low-Cost IoT Sensor for Smart Earthquake Detection and Notification.", KIBIM, Korea, 2020
- **Junyoung Park**, Junsik Shin, Jongbin Won, Jong-Woong Park, Minyoung Park, "Cloud based Bridge Displacement Monitoring System", COSEIK, Korea, 2021
- Suleman Khan, Jongbin Won, Junsik Shin, **Junyoung Park**, Jong-Woong Park, "A Low Power Strain Visualization System with Cloud-Based Data Management", KSMI, Korea, 2021
- **Junyoung Park**, Junsik Shin, Jongbin Won, Jong-Woong Park, Yun Jang, "Estimation of Bridge's Damaged Cable Using Deep Learning", KSMI, Korea, 2021
- Junsik Shin, **Junyoung Park**, Jongbin Won, Jong-Woong Park, "Cloud-based Long-term Bridge Monitoring System", KSMI, Korea, 2021
- Sehwa Yang, Gunhee Kim, Junsik Shin, **Junyoung Park**, Jong-Woong Park, "Cloud-based 3D Bridge Bearing Displacement Measurement using an IoT Sensing System", KSMI, Korea, 2021
- **Junyoung Park**, Junsik Shin, Sohaeng Kang, Jongbin Won, Jong-Woong Park, "Cloud-based Long-term Bridge Monitoring System", KSMI, Korea, 2021
- Sadia Umer Khayam, Jongbin Won, Junsik Shin, **Junyoung Park**, Jong-Woong Park, "IoT Sensing System for Monitoring Precast Structures During Delivery", KSMI, Korea, 2021
- **Junyoung Park**, Junsik Shin, Jong-Woong Park, Minyong Park, "Optimization of Acceleration-Strain based Displacement Estimation Method Considering Bridge Boundary Conditions", COSEIK, Korea, 2022
- Jongbin Won, **Junyoung Park**, Junsik Shin, Rayeon Kim, Jong-Woong Park, "Ultra-Low Power Strain Visualization Sensor and Chatbot-based Data Management System", COSEIK, Korea, 2022
- Jaehong Lee, Rayeon Kim, **Junyoung Park**, Junsik Shin, Jong-Woong Park, "Monitoring System for Prefab Structures during Transportation", KSNVE, Korea, 2022
- **Junyoung Park**, Jong-Woong Park, Junsik Shin, "Optimization of Acceleration-Strain based Displacement Estimation Method Considering Bridge Boundary Conditions", KSNVE, Korea, 2022
- Jong-Woong Park, **Junyoung Park**, Junsik Shin, Jongbin Won, "Cloud-based IoT Sensor System for Load Carrying Capacity of Bridges", KSPHM, Korea, 2022
- Jong-Woong Park, Rayeon Kim, Junsik Shin, **Junyoung Park**, Jongbin Won, "Ultra-Low Power Strain Visualization Sensor and Chatbot-based Data Management System", KSPHM, Korea, 2022
- **Junyoung Park**, Junsik Shin, Hyejung Jang, Jong-Woong Park, "Development of IoT Sensor System and Web-based Documenet Automation for Bridge Load-Bearing Capacity Evaluation", CONVR, Korea, 2022
- Junsik Shin, **Junyoung Park**, Jong-Woong Park, "Developmente of Cloud-AI based High-Sensitivity Wireless Smart Sensor for Port Structure Monitoring", CONVR, Korea, 2022
- Jaehong Lee, **Junyoung Park**, Junsik Shin, Rayeon Kim, Jong-Woong Park, "A Fully Automated Monitoring System for Precast Structures During Various Stages of Delivery", CONVR, Korea, 2022

- **Junyoung Park**, Jong-Woong Park, Junsik Shin, "Bridge Monitoring System using IoT sensor and Cloud Server", KSNVE, Korea, 2023
- Jaehong Lee, **Junyoung Park**, Junsik Shin, Jong-Woong Park, "Online Monitoring System for Precast Concrete Elements during Transportation", ITC, Korea, 2023
- Junsik Shin, **Junyoung Park**, Jong-Woong Park, "A Wireless Vibration Monitoring System for Berthing Structures in Ports", ITC, Korea, 2023
- **Junyoung Park**, Junsik Shin, Jong-Woong Park, "Cloud-based Ambient Bridge Monitoring using Strain and Acceleration Measurements", ITC, Korea, 2023
- Kwanghyun Kim, **Junyoung Park**, Jaehong Lee, Jong-Woong Park, "Development of a Low-Power IoT Sensor integrated with Cloud Services for Long-term Freight Train Monitoring", ITC, Korea, 2023
- **Junyoung Park**, Jong-Woong Park, "IoT Sensor and Cloud Computing based Long-term Bridge Monitoring", KIBSE, Korea, 2023
- Minyoung Kim, Soyeon Kim, **Junyoung Park**, Jong-Woong Park, Kyongkyu Choi, "TCP/IP-based Sensor Framework for Continuous Vibration Monitoring of Structures", EESK, Korea, 2023
- **Junyoung Park**, Junsik Shin, Jong-Woong Park, "A Scalable Framework for Ambient Bridge Monitoring using IoT Sensor and Cloud Computing", KSMI, Korea, 2023
- Junsik Shin, **Junyoung Park**, Jong-Woong Park, "Development of Back Analysis Algoritms for the Development of Unified Management System of Earth Retaining Mehods", KSMI, Korea, 2023

Patents

- Jong-Woong Park, Jongbin Won, **Junyoung Park**, Junsik Shin, "IoT Sensor and Cloud Computing for Ambient Bridge Displacement Monitoring." Korea Patent (Serial# 1020220021506), (Feb, 18, 2022).
- Seunggyun Park, Junsik Shin, **Junyoung Park**, "PARTICIPATORY SMART CITY THROUGH SENSING AND DATA VISUALIZATION SYSTEM." Korea Patent (Serial# 1023892220000), (Sep, 24, 2021)

Volunteer Experience

Da Vinci Dream Discovery Program / ***Chung-Ang University* – Seoul, Korea** 2020

Arduino-Based Earthquake Sensor and Alert System
• Created an introductory video for the CEE Department as part of a high school career exploration program

Undergraduate Research Mentoring

Practical & Research Engagement (PRE) Program / ***Chung-Ang University* – Seoul, Korea**

Port Berth Monitoring using IoT Sensors / Name: Junsik Shin 2022
• Developed IoT Sensors for continuous smart port monitoring
• Designed a Raspberry Pi-based IoT hardware and server system

Cloud-based 6-DOF Measurement Management System Development / Name: Gunhee Kim 2022
• Assisted in the development of an efficient power management system using EDS
• Developed a relational database for displaying measured 6-DOF displacements from aruco marker

Prefab Monitoring during Delivery using IoT Sensors / Name: Jaehong Lee 2022
• Developed an automated precast concrete management platform at various stages using IoT sensor
• Enabled a cloud based real-time data analysis system which facilitated damage assessment and on-site evaluation with automatic reporting

Cloud-based Freight Train Monitoring System / Name: Kwanghyun Kim 2023
• Developed IoT sensors for monitoring freight train bearing damage